Chemical Reactions & Equations

Ch - 1

Activity 1.1

magnesium Ribbon is burnt

$2Mg + O_2 \rightarrow 2MgO$
magnesium Oxide

Q) Why do we rub magnesium ribbon with sandpaper.

Ans) To remove the layer of magnesium oxide formed on the surface.

Observation

White dazzling light on burning white ashes (MgO)

Activity 1.3

$Pb(NO_3)_2 + KI \rightarrow 2PbI_2 + KNO_3$

- Form lead iodide, Potassium Nitrate

Activity 1.3

$Zn + HCl \rightarrow ZnCl_2 + H_2 \uparrow$

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2 \uparrow$

-> Effervesence
-> pop sound

Q) List some charactistics of a chem reaction
Ans) Change in state
Change in colour
Change in gas
Change in temp.

Chemical Equations

- $H_2 + O_2 \rightarrow H_2O$
(g) (g) (g)

* Chem reac. are symbolic representat

- $Mg + O_2 \rightarrow MgO$ (solid)

$2Mg + O_2 \rightarrow 2MgO$

- $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2 \uparrow$

Q) Why do we need to balance Chem. eq.
Ans) To satisfy law of Conservation of mass

$Fe + H_2O \rightarrow Fe_3O_4 + H_2$

$3Fe + 4H_2O \rightarrow Fe_3O_4 + H_2$

Types OF Chem. Reac.

1) Combination -> $A + B = AB$

2) Decomposition -> $AB \rightarrow A + B$

3) Displacement -> $AB + C \rightarrow A + CB$

4) Redox

Activity 1.4

$CaO + H_2O \rightarrow Ca(OH)_2 + \Delta$

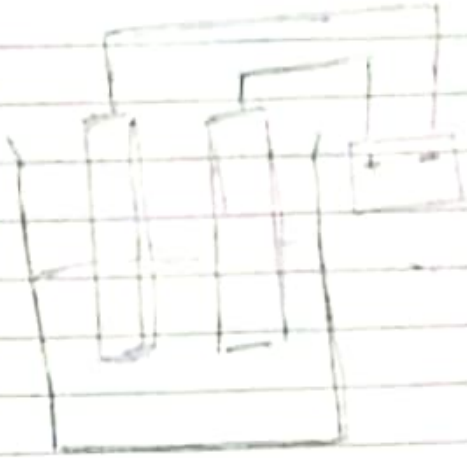
$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

Decomposition

- Thermal (Req heat)
- Electrolytic
- Photolytic

$FeSO_4 + 7H_2O \xrightarrow{\Delta} FeSO_4 + 7H_2O$

Electrolysis -> Split



H_2O

H_2 O_2

Displacement React.

Activity 1.9

Iron Nail

-> 20 mins

A $CuSO_4$ (Blue) B

$Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

$Cu + FeSO_4 \rightarrow$ No reaction

K	Fe	Au
Na	Sn	
Ba	Pb	
Ca	H	
Mg	Cu	
Al	Hg	
Zn	Ag	

Oxidation

$2Cu + O_2 \rightarrow 2CuO$

Copper (II) Oxide

$CuO + H_2 \rightarrow Cu + H_2O$

-> H oxidised to H_2O
CuO reduced to Cu.

Oxidation + Reduction

Redox

Rancidity

- Oxidation of fats & oils when left for long time, change smell & taste.

- **Prevention**

Put food in a tight container w. a lid.

Questions (NCERT)

A chem. eq. with equal number of atoms on both LHS & RHS is called a balanced chem. eq

- The chem. eq must be balanced in order to conserve the law of conservation of mass.

- Total mass of reactants must be equal to total mass of products.

$H_2(g) + N_2(g) \rightarrow NH_3$

$3H_2 + N_2 \rightarrow 2NH_3$

$H_2S + O_2 \rightarrow H_2O + SO_2$

$2H_2S + 3O_2 \rightarrow 2H_2O + 2SO_2$

$BaCl_2 + Al_2(SO_4)_3 \rightarrow AlCl_3 + Ba$

$3BaCl_2 + Al(SO_4)_3 \rightarrow 2AlCl_3 + 3BaSO_4$

$K + H_2O \rightarrow KOH + H_2$

$2K + 2H_2O \rightarrow 2KOH + H_2$

$HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + H_2O$

$2HNO_3 + Ca(OH)_2 \rightarrow Ca(NO_3)_2 + 2H_2O$

$NaOH + H_2SO_4 \rightarrow NaSO_4 + H_2O$

$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

$NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

$NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + HCl$

$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$

Calcium Hydroxide + Carbon Dioxide
→ Calcium Carbonate + Water.

→ $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$.

Zinc + Silver Nitrate → Zinc Nitrate
+ Silver

→ $Zn + AgNO_3 \rightarrow ZnNO_3 + Ag$.

Aluminum + Copper Chloride → Alum
Chloride +
Copper.

→ $2Al + 3CuCl_2 \rightarrow 2AlCl_3 + 3Cu$.

$Fe_2O_3 + 2Al \rightarrow Al_2O_3 + 2Fe$

-> Displacement Reac

$2KBr(aq) + 2BaI_2 \rightarrow 2KI + BaB_r$

-> Double Displacement Reaction

$ZnCO_3(g) \rightarrow ZnO + CO_2$

$H_2 + Cl_2 \rightarrow 2HCl$

$Mg + 2HCl \rightarrow MgCl_2 + H_2\uparrow$

Exothermic reactions -> Release heat

Endothermic " -> Absorb heat

eg -> $C + O_2 \rightarrow CO_2 + Heat$

$\rightarrow N_2 + O + Heat \rightarrow 2NO$

During respiration, we need energy. We get this energy from food we eat.

$C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + E$

This reaction is respiration.

Decomposition Reaction - Those in whi one compound breaks into more simple substanc

Combination Reaction - 2 or more substances combine to form a single compound.

Energy as heat

$2Pb(NO_3)_2 \xrightarrow{\Delta} 2PbO + 4NO_2 + O_2$

Energy as electricity

$2H_2O \longrightarrow 2H_2 + O_2$

Energy as light

$2AgCl \rightarrow 2Ag + Cl_2$

In Text Questions

magnesium ribbon is cleaned before burning so that coating of MgO is removed

$H_2 + Cl_2 \rightarrow 2HCl$

$3BaCl_2 + Al(SO_4)_3 \rightarrow 3BaSO_4 + 2A$

$2Na + 2H_2O \rightarrow 2NaOH + H_2$

Upmp 8.12.7.25